THALis: A Topology-Preserving Brain-Computer Interface for Dexterous Non-Invasive EEG-Driven Hand Control

CBIO039

Krishna Bhatt, BASIS Independent Silicon Valley, San Jose, California

INTRODUCTION

Core Problem: Over 80M people face loss of upper limb function.

- Non-invasive EEG signals offer **90% less signal quality** than invasive counterparts, limiting prosthetics to **binary control**

Challenges

- EEG signal is spatially smeared across the scalp
- Topology preservation fails under all traditional deep learning methods
- Prior deep learning decoders overfit to session-specific noise distributions

Research Objectives

1. Recovered **finger- level** low-dimensional manifolds via scalp EEG
2. Engineered a **topologically valid** decoder scheme
3. Achieved non-invasive **proportional**, multi-finger decoding

$100k Invasive Cost

40% Abandonment Rate

10:1 SNR Gap

PROCESS & PERFORMANCE VALIDATION

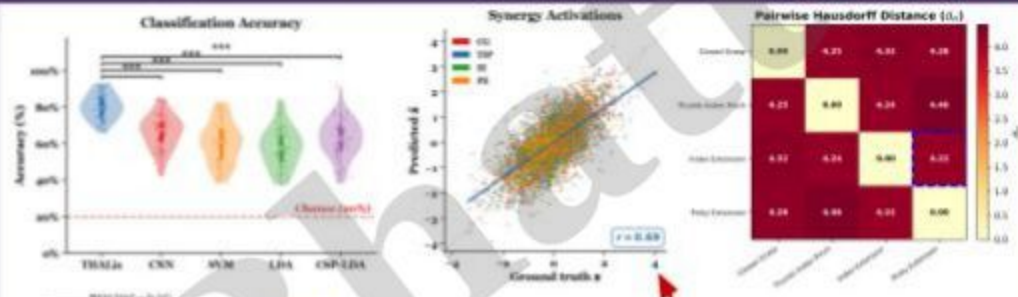
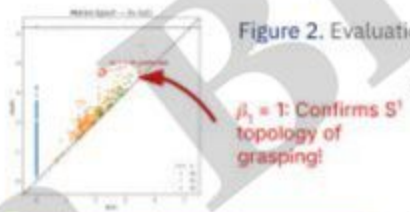


Figure 2. Evaluation results. Plots created by finalist with matplotlib, 2026.



54.5% improved tracking than CNNs

- The final system showed **significant improvement** on proportional motion tracking tasks ($p < 0.001$).
- Crucially demonstrates unique attractor separation across various primitives. (Cohen's d=1.19)

PROJECT FRAMEWORK

Core Innovation: THALis acts as a state-observer, overcoming the signal-to-noise bottleneck.

- The observer module: I designed a **bifurcating echo-state-network reservoir** to mimic cortical dynamics, conditioned on Riemannian separability and Hausdorff loss.

Based on the **Neural Manifold Hypothesis**:

- The neural state $\mathbf{x} \in \mathbb{R}^N$ can be modeled as $\mathbf{x} = f(\mathbf{z}) + \epsilon$, where $\mathbf{z} \in \mathbb{R}^M$ and $M << N$.

I derived various theorems, elucidating optimal parameter regimes for my novel system.

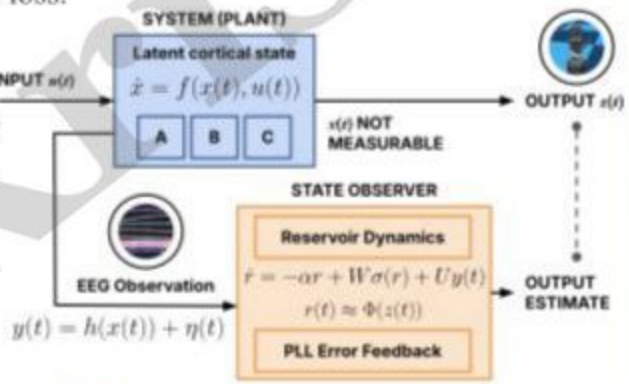


Figure 1. A state-observer perspective of THALis. Graphic created by finalist with Google Slides, 2026.

CONCLUSIONS AND IMPACT

- This work presents **the first non-invasive EEG system to produce proportional control**, achieving invasive-grade performance ($100k+) with noninvasive technology (< $1k).
- THALis opens a practical, highly accessible route towards restoring patients' quality of life



Figure 3. Picture taken by finalist, 2026.

- The topology-preserving echo-state reservoir is a **fundamental academic contribution** with broad applicability to any BCI.
- I established a theoretical and practical framework, laying **groundwork for future researchers**.

Unless otherwise noted, all figures, images, and data plots were created or taken by the researcher.

Krishna Bhatt